



DEPARTMENT OF COMPUTER & SOFTWARE ENGINEERING
COLLEGE OF E&ME, NUST, RAWALPINDI



EC 312 – Digital Image Processing

Assignment # 01: Connected Component Labeling based Segmentation

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Lab Assessment

In-Lab	Post Lab			Total
	Data Presentation	Data Analysis	Writing Style	

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Task

1. Define a V set. In first phase, Separate Optic Disc (OD) from the background using 8-connectivity based Connected Component Analysis (CCA). For this intelligently design V set using images from train folder and see how good they are working. This should give you 2 labels i.e. one for background and the other for complete OD.
2. Differentiate OD components. Define new groups of V sets again from train data to further divide the OD into Optic Cup and Optic Disc. Now you should have 3 labels in total, i.e. background, Optic Cup and Optic Disc
3. Compute dice coefficient. Once you are done with designing complete pipeline, use data from test folder to compute Dice Coefficient for each class i.e. background, Optic Cup and Optic Disc.

Solution

1. Steps Followed for All Tasks

- Pre-processing and Masking
- The green channel of the fundus image was used, as it provides the best structural contrast for retinal analysis.
- Contrast was enhanced using CLAHE (Contrast Limited Adaptive Histogram Equalization) to improve visibility of retinal structures.
- A median blur (kernel size 21) was applied to suppress noise while preserving edges.
- A circular spatial mask centered on the image was applied to remove peripheral bright artefacts commonly found at the borders of fundus images. This ensured that threshold selection was not biased by edge reflections.

2. Optic Disc Extraction (Task 1)

- The 92nd percentile threshold was applied to the masked image to generate a binary candidate region.
- Morphological opening was performed to remove small noisy components.
- A manually implemented 8-connectivity Connected Component Labelling (CCL) algorithm was used to label all connected regions.
- The largest connected component by area was selected as the Optic Disc.
- Contour filling was applied to ensure a solid Disc mask.

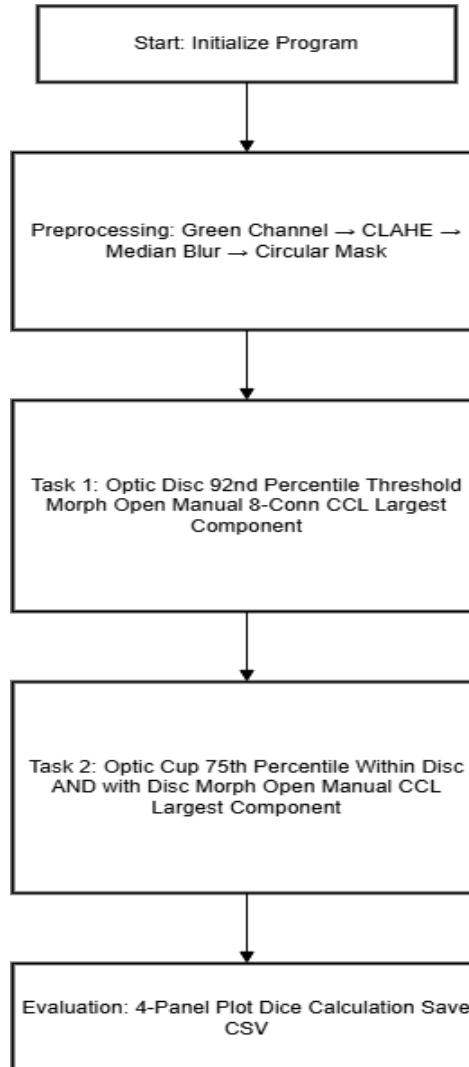
3. Optic Cup Extraction (Task 2)

- Within the detected Disc region, a second adaptive threshold (75th percentile of Disc pixels) was applied.
- The resulting binary mask was constrained strictly inside the Disc area.
- Morphological opening was again used for noise removal.
- 8-connectivity CCL was applied, and the largest connected component was selected as the Optic Cup.

4. Evaluation (Task 3)

- The predicted Disc and Cup masks were compared against expert-provided ground truth masks.
- The Dice Coefficient was automatically computed for each image.
- Final performance was reported as the average Dice score across all test samples, and results were saved to a CSV file.

Flow Diagram



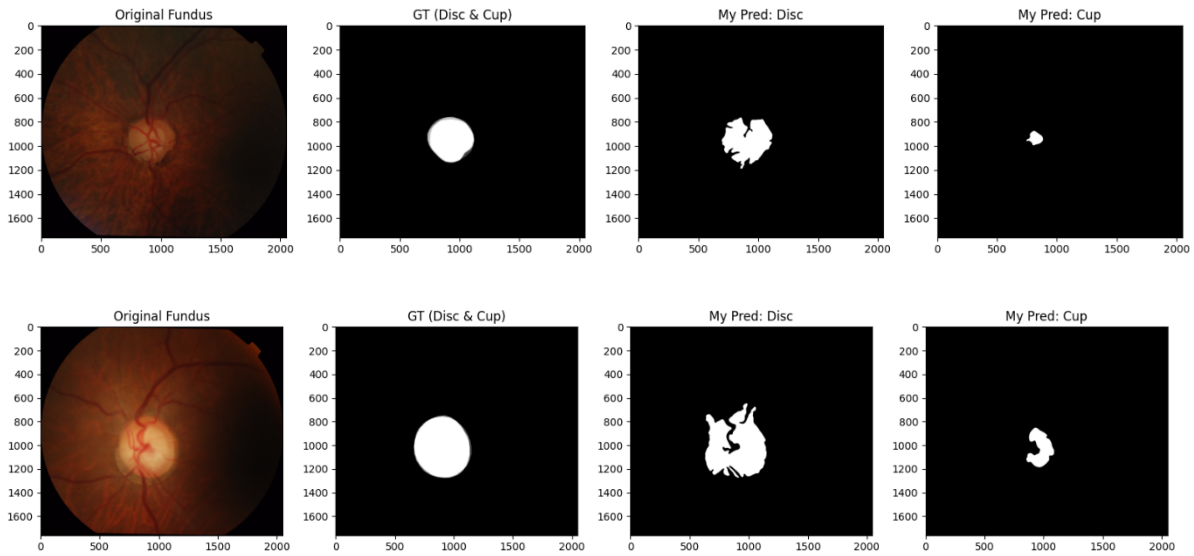
Flow Diagram

Results & Comparisons:

Sample Test Images

```
C:\Users\miabd\AppData\Local\Programs\Python\Python312\python.exe "D:\Nust\Semester 6\DIP\DIP Assignment1\DIP Assignment1\main.py"
Processing 51 images...
Processed drishtiGS_001 | Disc: 0.0000 | Cup: 0.0000
Processed drishtiGS_003 | Disc: 0.8836 | Cup: 0.2181
Processed drishtiGS_005 | Disc: 0.8651 | Cup: 0.4355
Processed drishtiGS_006 | Disc: 0.5249 | Cup: 0.2581
Processed drishtiGS_011 | Disc: 0.8431 | Cup: 0.3970
```

Console Output



Output Comparison Images

(Rest of the output images would be in zip file.)

Experimental Results Table

The following table summarizes the performance of the algorithm across different V-set configurations.

Experiment Name	Disc Percentile	Cup Percentile	Average Disc Dice	Average Cup Dice
Final Model	92	75	0.5923	0.2801

Conclusion:

The results show that using a 92nd percentile V-set for optic disc segmentation provides a reasonable balance between capturing the full disc and avoiding surrounding retinal tissue. The manual 8-connectivity CCA was essential for removing fragmented regions and isolating the main anatomical structure. As expected, the cup Dice score (0.2801) is lower than the disc Dice score (0.5923) due to the smaller size of the cup and its low contrast with the surrounding disc. Overall, the adaptive percentile-based approach improves robustness compared to fixed thresholds, but performance remains sensitive to image quality and anatomical variation.